



MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. EXAMINATIONS - JANUARY 2023

SUBJECT: ECE 1071 / ECE-1071 - BASIC ELECTRONICS

BASIC ELECTRONICS [ECE 1071]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

- 1A) Obtain an expression for the drain current of MOSFET in terms of ' V_{GS} ', ' V_{DS} ', and the dimension of the device. Also, plot the VI characteristics of the device indicating all the salient features on it. (4)
- 1B) Plot the VI characteristics of a Zener diode. Explain the different types of regulations that can be found in a Zener regulator with a neat circuit diagram and relevant equations. (3)
- 1C) Calculate the static and dynamic resistance of a PN junction Germanium diode at a temperature of 50°C , when the applied voltage is 0.25V and reverse saturation current is $10\text{ }\mu\text{A}$. (3)
- 2A) A sinusoidal voltage of 130V peak-to-peak from the secondary of the transformer is applied to a half-wave rectifier, with a load resistance of $1\text{ k}\Omega$. If the cut-in voltage of the diode is 0.7V and the forward resistance of the diode is 30Ω , find the values of the following. i) DC and RMS values of the load current. ii) DC output power and AC input power. (4)
- 2B) Draw the Zener voltage regulator circuit. Given, the series resistance connected in the circuit is $2\text{ k}\Omega$ and the load applied is $4\text{ k}\Omega$. Input voltage is varied between $(31\pm 9)\text{ V}$. Zener breakdown voltage is 10V. Find the minimum and maximum value of current through the Zener. (3)
- 2C) With a neat circuit diagram, explain the working of a full wave bridge rectifier with a capacitor filter. Draw the corresponding input and output waveforms. (3)
- 3A) A (7,4) bit Hamming code received is 0100111 ($P_1P_2D_3P_4D_5D_6D_7$), considering even parity. Check for errors in the received code and if there is an error, write the corrected code, Also represent the corrected code in 8421 BCD code. (4)
- 3B) The inputs $V_1=60\mu\text{V}$ and $V_2=30\mu\text{V}$ is given to the non-inverting and inverting terminals of an OPAMP, respectively. The common mode gain is 40dB and CMRR is 60dB. Calculate the output voltage. Also, Determine the output voltage when $V_1=V_2=30\mu\text{V}$ is given as input. (3)
- 3C) Realize an op-amp inverting integrator with $RC=1\text{ second}$, and input is a step voltage of 2V. Determine the output voltage at $t=4\text{ second}$ and sketch the output waveform. (3)
- 4A) Simplify the following SOP expression using K-Map and implement using NAND gates only
 $f(A, B, C, D) = \sum m(0,1,4,5,6,7,9,11,15) + \sum d(10,14)$. (4)

- 4B) Realise a 3-bit up counter using the positive edge triggered JK flip flops. Draw the timing diagram for the same. (3)
- 4C) With a neat logic diagram and truth table, explain the working of SR flip-flop, implemented using only NAND gates. Mention the limitation of SR flip-flop. (3)
- 5A) Certain AM transmitter radiates 12kW of power with carrier unmodulated and 15.250kW of power when carrier is sinusoidally modulated. Calculate the modulation index. If another sine wave corresponding to 60% modulation is transmitted simultaneously, determine the total power radiated. (4)
- 5B) Write the equation for Frequency Modulated signal in time domain for a sinusoidal modulating signal. Determine the bandwidth and modulation index of FM signal required to transmit 12 kHz, 3V peak amplitude signal with frequency sensitivity 18 kHz/V using Carson's rule. (3)
- 5C) **Define sampling theorem. Consider an analog signal,**
 $x(t)=15\cos(15\pi t)+10\sin(100\pi t)+5\cos(300\pi t)$.
Determine the minimum sampling rate required to reconstruct the signal. (3)

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